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Rheology and printability of cement paste modified with filler from manufactured sand

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Abstract. Manufactured sand is increasingly used in concrete production. We study the effects of substituting parts of the binder with manufactured filler (particle size 0.04 - 0.250 mm) on the printability of cement paste experimentally and numerically.

An experimental program explored water-to-binder ratios of 0.4 and filler-to-binder mass ratios $f_i/b = 0.15 - 0.85$. The study investigated suspension properties and rheology like maximum packing, mini-slump, plastic viscosity, and yield stress. Printability, on a lab scale, was observed by printer flow measurements (“extrudability”) and buildability (shape retention, deformation layer adhesion, deformation, macro voids). Additionally, a computational fluid dynamics model (CFD) is developed to simulate the 3D printing of the filler-modified cement paste and assess its printability.

Results show that as maximum packing increases, slump flow (yield stress) decreases as expected. The best shape retention and layer adhesion were found in mixes with maximum packing = 0.94, 0.96, and 0.98. The mix with the highest yield stress showed the best shape retention but had more macro voids in the cross-section and rougher surfaces. To confirm the accuracy of the CFD model, the cross-sectional shapes of the deposited part from simulations are compared with the printed ones. It seems more work is needed to get a good correlation with the same process parameters.

Keywords: Printability, Yield stress, Packing, Manufactured filler, Mini-slump, CFD

1 Introduction

The advancement of 3D printing in construction, specifically in concrete and cementitious material printing, represents steps forward in the modernization of construction practices. Within this realm, the rheology and printability of filler-modified cement pastes have become significant areas of research when aiming to reduce the environmental impact associated with cement-rich 3D printing processes. Natural aggregate sources, such as glacio-fluvial sand, diminish and there is an increasing trend towards utilizing manufactured aggregates derived from local bedrock [1]. This shift affects transportation, environmental consequences, and the quality of concrete. Substituting cement with manufactured fillers could reduce cement consumption but also influence rheological characteristics such as yield stress and viscosity [2], which are crucial for defining the printability window of cementitious materials [3]. Incorporating manufactured filler into the paste while obtaining desirable rheological properties requires a well-controlled aggregate production in terms of crushing, shaping and particle size as well as skilled proportioning [2, 4, 5]. From experience, we know how particle size can be used to control filler surface area by composing (micro-proportioning) fillers and that the surface area effect correlates clearly to rheology, much more importantly than the mineralogy [1]. Moderate changes in filler particle shape [6] had a relatively weak effect on rheology when controlling size. A marked effect was observed on the rheology of filler modified paste for filler with 1.5 % absorption compared to in a filler with 0.1 % absorption.

The simulation, in addition to experiments on printability can be used to predict their performance under various extrusion and layering conditions in additive manufacturing processes. By comparing numerical simulations to printing experiments on materials with known rheological properties we could establish rheological criteria for materials intended for shaping.

Previous research utilizing Computational Fluid Dynamics (CFD) has enabled predictions about the effects of various rheology- and processing parameters on the end product's structural and geometrical attributes, such as the cross-sectional shape of the extruded strands that result from their deposition [7-10]. Furthermore, it was shown how variations in yield stress and plastic viscosity affect layer deformations. An increased yield stress leads to prints with fewer deformations within the deposited layers, whereas an increased plastic viscosity results in greater deformations because the extrusion pressure increased at the nozzle outlet when the material's resistance to flow increased [11]. Additionally, the significance of yield stress buildup in preventing deformation was highlighted [12].

The addition of manufactured fines changes the rheological properties and printability of cement paste. Our experience so far has mainly been with flowable / highly flowable filler modified cement paste. It is therefore of interest to investigate how flowable paste can be further modified into paste suitable for printing by using fillers from manufactured sand. Furthermore, it is of interest to investigate the possibilities and potential to reduce the high cement content often seen in printable concrete [3] with filler of manufactured sand.

This paper is an experimental and numerical study of the rheological behavior and printability of cement paste modified with filler from manufactured sand.

2 Methodology

2.1 Materials

Fillers in this project were produced via a three-step crushing process, ending with a Vertical Shaft Impactor optimizing particle shape, followed by mechanical and air classification [1]. The filler-modified cement paste, or the matrix, includes limestone manufactured fillers (T5) with size ranges 0.04 mm to 0.25 mm. The mineralogical compositions and physical properties of T5 are detailed in references [1]. Additionally, Norwegian cement (CEM I 52.5 R), Fly Ash, and silica fume 940-U are utilized. Dynamon SR-N, with a dry solids content of 17.5%, is used as a superplasticizer. Figure 1 shows the volume fractions of matrix components. The measured densities are 1909, 1995, and 2085 kg/m³ for matrix 1, 2, and 3 respectively.

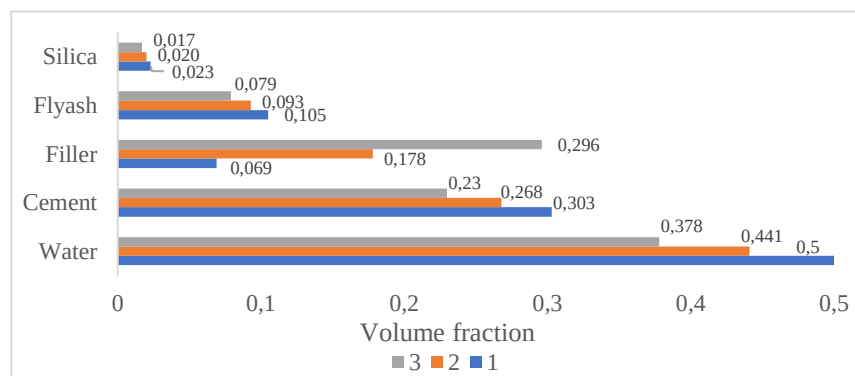


Fig.1. volume fractions of filler-modified cement paste components. In all the mixtures, silica to binder mass ratio (s/b) = 0.04, and fly ash to binder mass ratio (FA/b) = 0.2, and superplasticizer/ c = 0.01

2.2 Experiments

Flowability was measured using a mini-slump cone (big diameter: 89mm, small diameter: 39mm, and height: 70mm). The mini-slump test involved filling a cone with freshly mixed cement paste and then slowly lifting the cone to let the paste flow under its own weight. The maximum packing of solids in the matrix was determined by centrifugation [13]. The rheological parameters were measured by an Anton Paar Physica MCR 300 rheometer. The measurements were performed using rotational controlled shear rate (CSR) tests with a vane-in-cup geometry (with a diameter of 22mm, length of 16mm,

and inner cup diameter is 28.92mm), chosen to minimize the risk of wall slip during the measurements and ensure accurate characterization of the material's bulk properties [14]. A pre-shearing phase was included, involving 60 seconds of pre-shear (30 seconds at a shear rate of 60 s^{-1} followed by 30 seconds of rest). This phase eliminates the initial reversible structuring of the mortar, ensures the sample is homogenized, and simulates the shear deformation that occurs during the printing process [10].

Following the ascending part by increasing the shear rate from 0 to 60 s^{-1} , and descending by decreasing the shear rate from 60 to 0 s^{-1} . The printability of the matrix was evaluated by extruding multi-layer beams with a HYREL 3D Printer, Engine SR model, on a small scale. The orifice of the nozzle is rectangular, with a length of 20 mm and a width of 4 mm. With the help of the G-code, the length and height of each printed layer are set to 90 mm and 3.5 mm respectively, with a total of six layers being continuously printed at a uniform travel speed (V) of 5 mm/s to construct each specimen. The printer is shown in Fig. 2a. Also, a balance was added to the printer setup to measure the mass flow for different matrix, (Fig. 2b).

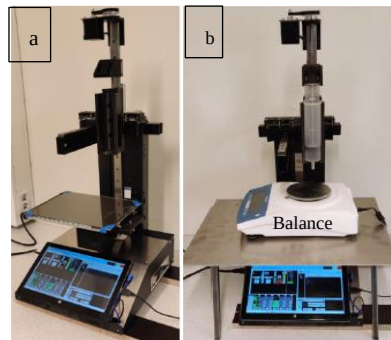


Fig.2. a: 3D printer with print head SDS 150 connected, b: stand and balance used to measure how much of matrix printed per second

2.3 Simulation

A CFD model is developed to simulate the deposition of an incompressible, filler-modified cement paste with a specific density. The model comprises a rectangular extrusion nozzle and a solid substrate. The dimensions of the nozzle orifice, the gap between the nozzle outlet and solid substrate, and the printing speed are the same as in the experiments. The extrusion rate of the material (U) is 7 mm/s, which is calculated based on the mass flow through the balance and cross-section of the nozzle [7]. According to the HYREL 3D Printer specifications, the material is extruded through a rectangular nozzle with the same dimensions as in the above lab experiment onto a solid substrate. The speed ratio, the ratio of printer speed to material speed, is $V/U=0.7$.

The geometry of the model is meshed with the Cartesian grid. The computational domain is assigned a symmetry boundary condition along its length, allowing half of

the domain to be solved. The bottom plane is the solid substrate with the boundary condition of a stationary wall while the top plane is assigned to an inlet boundary condition with the constant extrusion speed. The other planes have been assigned a continuative boundary condition, which presupposes that the velocity field's derivative is zero at the boundary surface, and these surfaces do not interact with the material being extruded [11].

The flow dynamics is governed by the Navier-Stokes equations:

$$\begin{aligned}\nabla \cdot \mathbf{u} &= 0 \\ \rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) &= -\nabla p + \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}\end{aligned}\quad (1)$$

where $\mathbf{u}$ is the velocity vector field, p is the pressure, ρ is the material density, $\mathbf{g}$ is the gravity acceleration vector, t is the time, and $\boldsymbol{\sigma}$ is the deviatoric stress-tensor.

The deviatoric stress-tensor is approximated by the generalized Newtonian fluid model:

$$\boldsymbol{\sigma} = 2\mu_{app}(\dot{\gamma}) \mathbf{D}_T \quad (2)$$

where μ_{app} is apparent viscosity, $\dot{\gamma} = \sqrt{2\text{tr}(\mathbf{D}_T^2)}$ is the strain rate and $\mathbf{D}_T = ((\nabla \mathbf{u}) + (\nabla \mathbf{u})^T)/2$ is the deformation rate tensor.

The viscosity of filler-modified cement paste is modeled with the bi-viscous regularization of the Bingham rheological model:

$$\mu_{app}(\dot{\gamma}) = \begin{cases} \mu_{m.app}, & \text{for } \dot{\gamma} < \dot{\gamma}_c \\ \mu_p + \frac{\tau_0}{\dot{\gamma}}, & \text{for } \dot{\gamma} \geq \dot{\gamma}_c \end{cases} \quad (3)$$

where, μ_p is plastic viscosity, $\mu_{m.app}$ is the maximum apparent viscosity, $\dot{\gamma}_c$ is critical shear rate, and τ_0 is yield stress.

The model, as described by Equations (1), is computationally solved using the FVM through the CFD software **FLOW-3D**[®] [9]. A semi-implicit method is employed for iterative calculations to determine the fields related to pressure and velocity. Moreover, the Volume-of-Fluid (VOF) approach was used in monitoring the free surface of the cementitious mixture. Simultaneously, the immersed boundary technique was utilized to simulate the nozzle's actions and its positioning within the computational domain.

3 Results and discussions

3.1 Printability

According to Fig. 3, captured during the printing process, the filler-modified paste with a Filler/b ratio of 0.15 exhibits a softer print compared to the paste with a Filler/b ratio of 0.85, which displays a stiffer printed consistency.



Fig. 3. Printed sample for w/b=0.4 and sp/c=1%

3.2 Rheological test

The rheological behavior of the filler-modified cement paste was simplified by applying the Bingham model to the data within a shear rate interval from 2 to 60 s^{-1} , for those mixes with w/b=0.4.

The pre-shearing phase is used to break down the initial, temporary structure in the matrix, and make it more uniform throughout the rheological measurement. Therefore, after the pre-shear, the peak is used to calculate the static yield stress, and the descending part for dynamic yield stress and plastic viscosity (seen in Table 1).

Based on Table 1, both static and dynamic yield stresses increase with the filler content. The plastic viscosity for all mixes grew as the filler replacement increased. This trend suggests that the addition of more filler particles enhances the paste's resistance to initial and continuous flow. As found in previous research [4, 5], the differing adsorption characteristics of superplasticizers, the particle size distribution, and the packing of particles and solid volume fraction of cement pastes could influence the variation of yield stress and viscosity likely due to the interparticle contact interaction, particle flocculation, and particle friction contacts. Accordingly, increasing the replacement of cement by manufactured filler hinders the movement within the paste matrix, so plastic viscosity and yield stress show an increasing trend with an increase in filler content.

Table 1: Rheological properties of matrixes

f_i/b	τ_{Static} (Pa)	τ_{Dynamic} (Pa)	μ_p (Pa.s)	R^2	Maximum packing	Slump flow (mm)	Predicted τ (Pa)[15]*
0.15	17.7	13.2	0.09	0.98	0.94	134	158.4
0.44	60.4	37.8	0.23	0.97	0.96	110.5	435.4
0.85	72.9	50.3	0.35	0.98	0.98	94.5	998.2

*The model is practical for slump flow higher than 200mm.

The yield stress and slump flow are correlated, indicating that an increase in yield stress results in a decrease in slump flow. The predicted yield stress is based on the model presented in the reference [15]. Although this model is compatible with flowable paste, it is observed that the yield stress of the matrix increases with the replacement of cement with more filler. The higher measured yield stress values for mixtures with higher f_i/b ratios align with the observed printability characteristics, where these mixtures showed better shape retention and less deformation compared to the lower f_i/b ratio. This is better illustrated in Figure 5, which shows the cross-section after hardening.

3.3 Simulation results

A graphical representation of the printing layers as shown in Fig. 4, illustrates the simulated layers. The static yield stress and plastic viscosity used in the simulation are according to Table 1. Figure 4 shows that for $V/U=0.7$ the model just simulated the first layer and diverged due to an increase of extrusion pressure at the nozzle outlet. This is because an excess amount of material accumulates around the extrusion nozzle when the extruder continues to push out material at a high rate while the movement of the nozzle is slow (i.e., speed ratio smaller than one). Previously, the significance of the V/U ratio was studied, demonstrating its influence on the geometry of the printed layers [10]. We increased the V/U ratio for multilayer printing optimization. A V/U ratio of 1 allows for successful multilayer printing in the simulation.

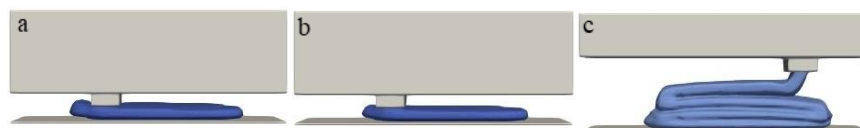


Fig. 4. a) $V/U=0.7$ led to print only one layer, b) $V/U=0.8$ led to print only one layer, c) $V/U=1$ led to print multilayer

Figure 5 shows the cross-sectional shapes of the 3D printed and simulated parts. Both experiment and simulation captured that the higher plastic viscosity and yield stress in $\text{Filler}/b=0.85$ and 0.44 resulted in less deformation compared to $\text{Filler}/b=0.15$. Furthermore, the simulated cross-sectional shape obtained for the case with $\text{Filler}/b=0.15$ has a good correlation with the experiment, whereas the other cases show some discrepancies. Despite the fact that the simulations were carried out with a smaller number of layers than the experiments, the discrepancies can be attributed to the fact that the model does not consider early setting or time dependent yield-stress build-up during the printing, which could be considered in future work. Furthermore, the CFD model can be extended to solve the elastic stress of the paste using an Elasto-viscoplastic model, which could offer better predictability. Overall, the CFD platform offers a new venue for investigating 3D printing with filler-modified cement paste.

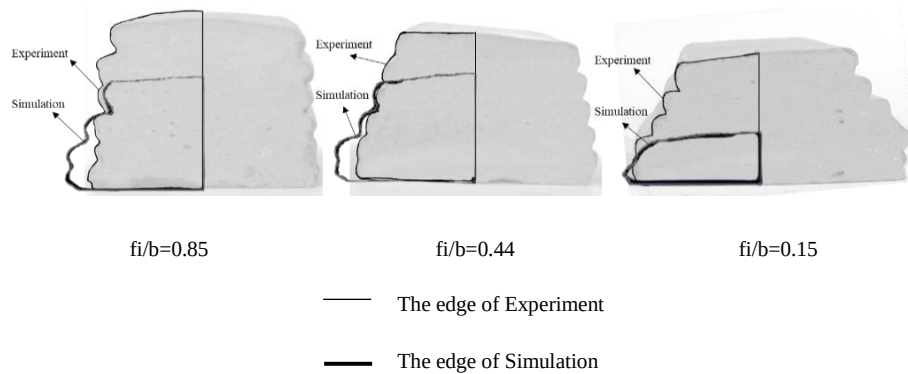


Fig. 5. Cross-section shapes of CFD model and experiments

As shown in Fig. 5, there is a good correlation between the cross-sectional shapes of the simulated and actual prints (experimental results). This indicates that the chosen CFD model with $V/U=1$ is likely capable of simulating the 3D printing process. The differences among the three shapes in Fig. 5 show the model's sensitivity to the application of different yield stresses and viscosities. Higher viscosity and yield stress in $fi/b=0.85$ and 0.44 resulted in less deformation compared to $fi/b=0.15$. It should be noted that the simulation was conducted for only 2 to 4 layers. Therefore, developing this model leads us to a model that closely mimics 3D printing process which is crucial for making accurate predictions of the printed sample.

4 Conclusion

Experimental and numerical studies of the rheology and printability of filler-modified cement paste showed that increasing the replacement of cement with filler led to higher maximum packing values (0.94, 0.96, and 0.98) and increases in yield stress and viscosity, while the slump flow decreased. A higher slump flow corresponds to a more fluid and easily flowing paste. As yield stress and viscosity increase due to higher filler

content, the paste's ability to flow freely under its weight decreases. The study highlights the importance of the speed ratio in achieving multilayer printing. A speed ratio (V/U) of 1 resulted in successful simulations for multilayer printing. Additionally, increasing the yield stress and viscosity in the CFD model revealed different deformations relevant to the rheology of the filler-modified cement paste.

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